seed_configs parameter table

Group	Parameter	Description	Typical / Default
Grid	u0	Initial u; 0 means start from initial data.	0; restart runs use 1.28
Grid	umax	Stop u-direction evolution here.	3.6 for coarse runs; 3.3 for zoom runs
Grid	vmax	Stop v-direction evolution here.	40 for coarse runs; 5 for zoom runs
Grid	h	Grid spacing; smaller = finer but slower.	0.01 / 0.001 / 0.0005 / 0.00025
Grid	dv	v-direction spacing; 0 means equal to h.	0
Grid	first_step_factor	Step-size scale for the first step.	1.0
Grid	r_floor	Minimum allowed r; prevents division by zero.	2e-5 or 1e-12
Grid	singularity_cut	Singularity cut-off threshold.	5e-3 or 1e-12
Grid	vacuum_r_threshold	Vacuum-region detection threshold.	0.0
Grid	ru_max	Upper bound on r_u.	1e8
Grid	rv_max	Upper bound on r_v.	1e8
Grid	alpha_floor	Lower bound on α.	1e-12
Grid	alpha_ceiling	Upper bound on α.	1e6
Grid	corrector_iterations	Number of corrector iterations per step.	1
Grid	center_iterations	Max center iterations.	20
Grid	center_tolerance	Center iteration tolerance.	1e-8
Grid	center_taylor_order	Taylor expansion order at the center.	2
Grid	center_damping	Center iteration damping.	0.5
Grid	center_fit_min_offset	Minimum offset for center fit.	1
Grid	center_fit_n_points	Number of points used in center fit.	8
Grid	center_fit_radius_scale	Radius scale for center fit.	3.0
Grid	center_project	Whether to apply parity projection at the center.	False
Grid	center_project_cells	Number of cells projected.	6
Grid	center_project_mode	Projection mode.	block
Grid	use_numba	Whether to use numba acceleration.	True
Grid	use_center_sources	Whether to use center source terms.	False
Grid	center_alpha_filter	Center α filter (experimental).	False
Grid	center_alpha_mode	Center α mode.	fit
Grid	center_scheme	Center scheme: ode or legacy.	ode
Grid	restart_file	Saved directory to restart from; empty = start from initial data.	empty; or parent directory name
Grid	restart_u	Row u of the parent solution to take initial data from.	1.28
Grid	restart_tag	Parent solution file prefix.	base_coarse / pert_coarse / single

Group	Parameter	Description	Typical / Default
Grid	center_block_cells	Number of cells in center block.	40
Grid	center_block_x0	Physical size of center block; 0 = by cell count.	0.0
Grid	freeze_alpha	If > 0, freeze α evolution.	0
Initial data	L0	Overall length scale of the initial data.	1.0
Initial data	family	Shape family of the initial data.	bump (all cases here use bump)
Initial data	amplitude	Overall amplitude a; larger values make horizon formation easier.	around 3.0
Initial data	epsilon	Perturbation strength ε; 0 is the baseline, ± are perturbations.	0, ±1, 0.5, 2
Initial data	xc	Pulse center position.	2.0
Initial data	width	Pulse width.	0.6
Initial data	xc2	Second pulse center.	2.2
Initial data	width2	Second pulse width.	0.8
Initial data	deformation_scale	Deformation scale for shape family; derived quantity.	≈ 0.213
Initial data	tp_a1	Two-pulse baseline weight 1.	1.0
Initial data	tp_a2	Two-pulse baseline weight 2.	0.55
Initial data	tp_b1	Two-pulse perturbation weight 1.	-0.35
Initial data	tp_b2	Two-pulse perturbation weight 2.	1.0
Initial data	x_max_norm	Upper limit of the norm integral.	12.0
Initial data	ell_x	Length scale of the norm.	1.0
Initial data	norm_points	Number of points in the norm integral.	24001
Initial data	norm_weight_power	Power exponent of the norm weight.	2.0
Initial data	norm_weight_amplitude	Amplitude of the norm weight.	0.25
Analysis	lambda_star	Matching point λ*; selects the horizon segment used for comparison.	0.3 (paper figures/tables use 0.3)
Analysis	horizon_transversality_min	Lower bound on horizon transversality.	1e-4
Analysis	min_horizon_points	Minimum number of horizon points.	6
Analysis	min_curve_points	Minimum number of points on the matched curve.	6
Analysis	rho_max_fit	Upper bound of the fitting window.	0.35
Analysis	rho_min_fit	Lower bound of the fitting window.	1e-3
Analysis	savgol_window	Smoothing window size.	11; figure scripts commonly use 7
Analysis	savgol_order	Smoothing polynomial order.	3
Analysis	kasner_tolerance	Tolerance for the Kasner criterion.	0.15
Analysis	equal_mass_tolerance	Equal-mass tolerance.	5e-4
Analysis	mass_root_span	Bracketing interval for the equal-mass root.	0.15
Analysis	fit_nested_windows	Number of nested fitting windows.	5
Analysis	match_scheme	How the two curves are matched.	equal_rho

Group	Parameter	Description	Typical / Default
Analysis	outer_mass_rmin	Lower bound of the outer-mass integral.	6.0
Analysis	outer_mass_rmax	Upper bound of the outer-mass integral.	40.0
Analysis	fixed_kasner_window_lo	Lower bound of the fixed Kasner window.	1e-3
Analysis	fixed_kasner_window_hi	Upper bound of the fixed Kasner window.	1e-1
Analysis	minkowski_compare_r_min	Lower bound of the Minkowski comparison.	1.0
Analysis	minkowski_compare_r_max	Upper bound of the Minkowski comparison.	5.0
Analysis	minkowski_margin	Boundary margin for Minkowski comparison.	3
Analysis	minkowski_pass_threshold	Minkowski residual threshold.	1e-8
Analysis	constraint_r_floor_factor	Constraint r lower-bound factor.	100.0
Analysis	constraint_boundary_margin	Boundary margin for constraint evaluation.	4
Analysis	constraint_r_max_fraction	Constraint r upper-bound fraction.	0.7
Analysis	constraint_r_min_fixed	Fixed lower bound for constraint evaluation.	0.5
Analysis	constraint_r_max_fixed	Fixed upper bound for constraint evaluation.	5.0
Analysis	constraint_conv_r_min	Lower bound of convergence test.	0.5
Analysis	constraint_conv_r_max	Upper bound of convergence test.	5.0
Analysis	constraint_conv_n_points	Number of points in convergence test.	40
Analysis	massflux_r_floor_factor	Mass-flux r lower-bound factor.	20.0
Analysis	massflux_horizon_fraction	Mass-flux horizon fraction.	0.9
Analysis	invalid_mass_penalty	Penalty for invalid mass.	1e6
Analysis	constraint_include_normalized	Whether to output normalized constraints.	True

Actual parameters for each computation case

No.	Phase	Case	tag	h	restart parent / tag@u	Domain u×v	family	a / ε	Initial data source
1	1	P1_01_eqzoom_base	base_coarse	0.01	—	3.6×40	bump	3 / 0	Initial data
2		P1_02_eqzoom_pert	pert_coarse	0.01	—	3.6×40	bump	2.86517 / 1	Initial data
3		P1_03_scan_eps-1_pert	pert_coarse	0.01	—	3.6×40	bump	3.13293 / -1	Initial data
4		P1_04_scan_eps0.5_pert	pert_coarse	0.01	—	3.6×40	bump	2.93282 / 0.5	Initial data
5		P1_05_scan_eps2_pert	pert_coarse	0.01	—	3.6×40	bump	2.72835 / 2	Initial data
6		P1_06_full_small_base	base_coarse	0.01	—	3.6×40	bump	3 / 0	Initial data
7		P1_07_full_small_h005_base	base_coarse	0.005	—	3.6×40	bump	3 / 0	Initial data
8		P1_08_family_bump_eps0.5_base	base_coarse	0.01	—	3.6×40	bump	3 / 0	Initial data
9	2	P2_01_zoom_base_h1e3	single	0.001	eqzoom_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
10		P2_02_zoom_pert_h1e3	single	0.001	eqzoom_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
11		P2_03_zoom_bump_eps0.5_base_v3	single	0.001	family_bump_eps0.5_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
12		P2_04_zoom_eps-1_pert_v3	single	0.001	scan_eps-1_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
13		P2_05_zoom_eps0.5_pert_v3	single	0.001	scan_eps0.5_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
14		P2_06_zoom_eps2_pert_v3	single	0.001	scan_eps2_v2 / pert_coarse @1.28	3.3×5	bump	2.7204 / 2	restart
15		P2_07_zoom_base_h5e4b	single	5.0e-04	eqzoom_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
16		P2_08_zoom_pert_h5e4b	single	5.0e-04	eqzoom_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
17		P2_09_zoom_bump_eps0.5_h5e4	single	5.0e-04	family_bump_eps0.5_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
18		P2_10_z5_eps-1	single	5.0e-04	scan_eps-1_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
19		P2_11_z5_eps0.5	single	5.0e-04	scan_eps0.5_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
20		P2_12_z5_eps2	single	5.0e-04	scan_eps2_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
21		P2_13_h4_v5_x2_base	single	2.5e-04	eqzoom_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
22		P2_14_h4_v5_x2_pert	single	2.5e-04	eqzoom_v2 / pert_coarse @1.28	3.3×5	bump	2.86517 / 1	restart
23		P2_15_zoom_bump_eps0.5_h25e5	single	2.5e-04	family_bump_eps0.5_v2 / base_coarse @1.28	3.3×5	bump	3 / 0	restart
24		P2_16_h25_eps-1	single	2.5e-04	scan_eps-1_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
25		P2_17_h25_eps0.5	single	2.5e-04	scan_eps0.5_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart
26		P2_18_h25_eps2	single	2.5e-04	scan_eps2_v2 / pert_coarse @1.28	3.3×5	bump	3 / 0	restart